

Support Vector Machine (Lab)

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2025-11-15

Table of contents

1	Loading Packages	3
2	Loading the data	4
3	View the data	5
4	Prepare for Trainig	7
5	Train the SVM - Linear kernel	8
5.1	Predictions	9
5.2	Results	10
5.3	Overfitting?	11
6	Train the dataset on radial kernel	12
6.1	Plots	12
6.2	Predictions	13
6.3	Results	14
6.4	High cost ($C = \text{large}$)	14
6.5	Low cost ($C \text{ small}$)	15

1 Loading Packages

```
#  
library(tidyverse)  
library(e1071) # package for SVM  
library(caret) # helper functions
```

2 Loading the data

Inspect the structure of the data

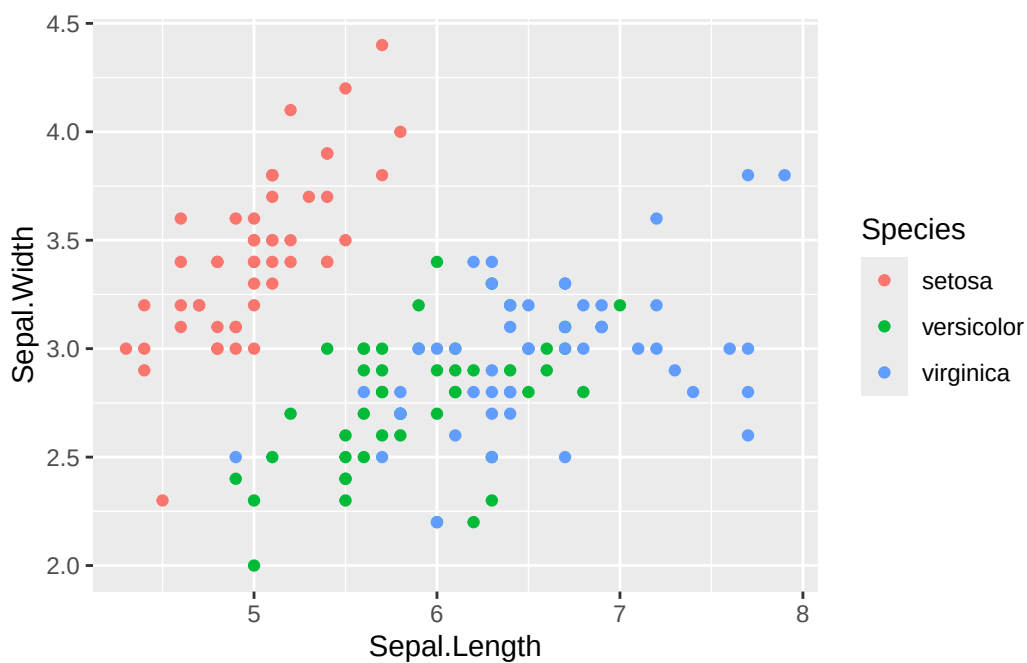
```
data(iris)
str(iris)
```

```
'data.frame':  150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1 ...
```

3 View the data

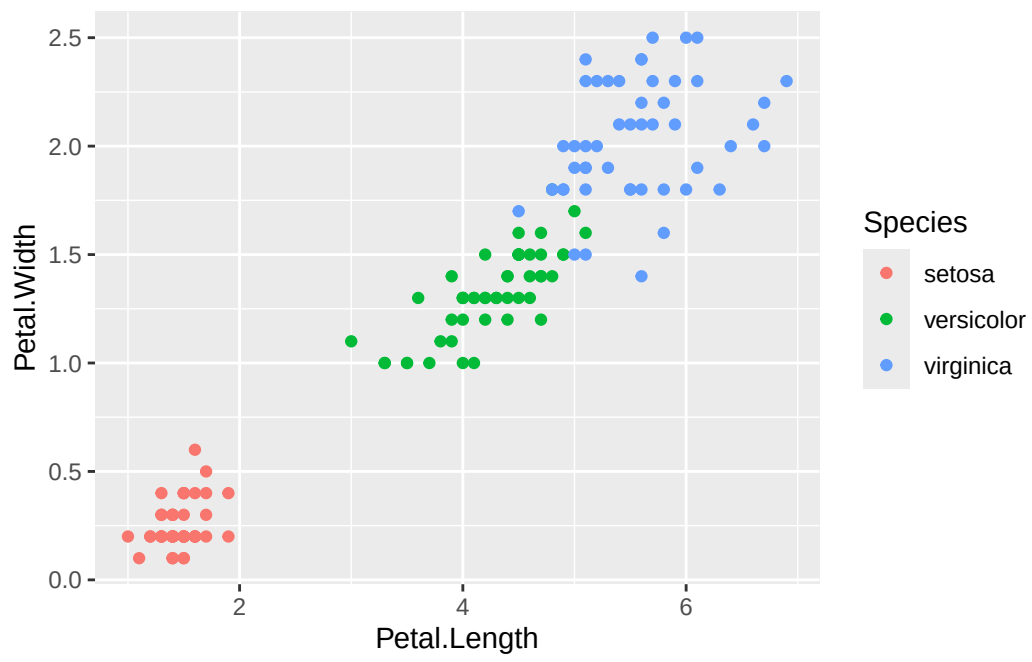
Plot the data by Sepal

```
iris |>  
  ggplot(aes(x = Sepal.Length, y = Sepal.Width, color = Species))+  
  geom_point()
```



Plot the data by petal

```
iris |>  
  ggplot(aes(x = Petal.Length, y = Petal.Width, color = Species))+  
  geom_point()
```



4 Prepare for Trainig

This function creates a stratified split of data. It splits the dataset into training and testing $p = 85$ (85% training) while preserving the class proportion of the Species variable. In other words this makes sure the proportion of each class (setosa, versicolor, virginica) in the split is the same as in the original dataset. List = FALSE - when you want the vector as a row numbers not as a list

```
set.seed(42)
indices <- createDataPartition(iris$Species, p = .85, list = FALSE)
```

Then I use it like this:

- train = 85% rows
- test_in = 15% (remainig) -indices
- test_truth = actual labels for evaluating predictions

```
train <- iris %>% slice(indices)
test_in <- iris %>% slice(-indices) %>% select(-Species)
test_truth <- iris %>% slice(-indices) %>% pull(Species)
```

5 Train the SVM - Linear kernel

The SVM function has the default cost of 10

```
set.seed(42)
iris_svm <- svm(Species ~ ., train, kernel = "linear", scale = TRUE, cost = 10)
summary(iris_svm)
```

Call:

```
svm(formula = Species ~ ., data = train, kernel = "linear", cost = 10,
     scale = TRUE)
```

Parameters:

```
  SVM-Type:  C-classification
 SVM-Kernel:  linear
       cost:  10
```

Number of Support Vectors: 17

```
( 2 8 7 )
```

Number of Classes: 3

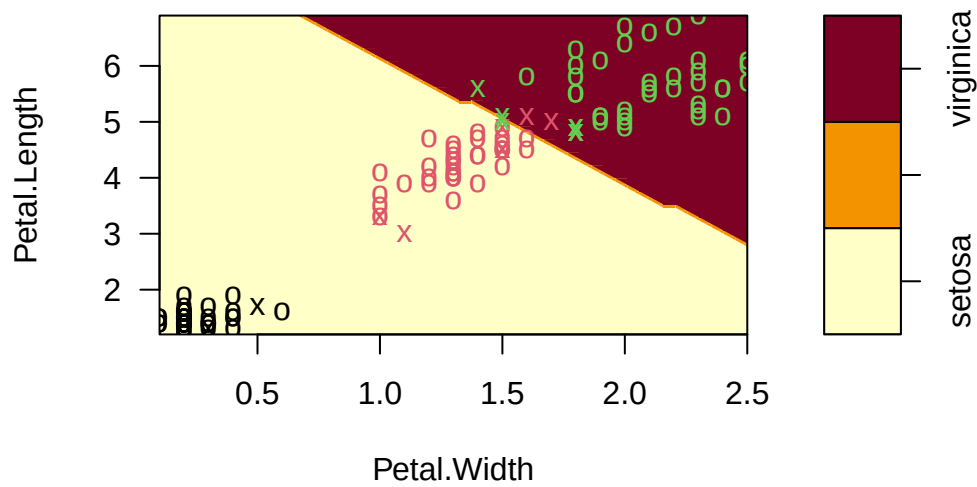
Levels:

```
setosa versicolor virginica
```

we can visualize the SVM decision boundaries only in two dimensions, even though the model was trained in four dimensions (all iris features).

```
plot(iris_svm, train, Petal.Length ~ Petal.Width)
```

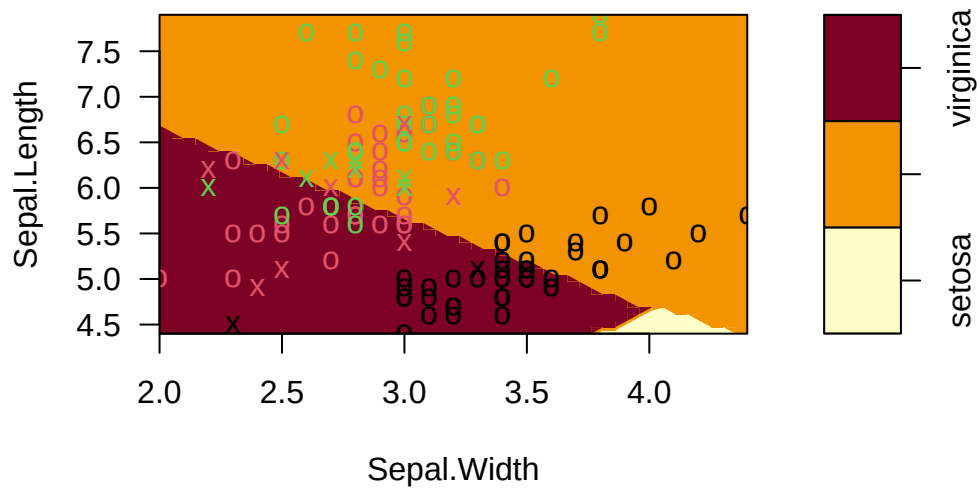

SVM classification plot



For Sepal leaf Dimensions it is needed to be sliced the other dimensions at a reasonable point

```
plot(iris_svm, train, Sepal.Length ~ Sepal.Width,
     slice = list(Petal.Length = 4.5, Petal.Width = 1.75))
```

SVM classification plot



The plots does not show the full SVM, only one projection at the time of the decision Surface into two dimensions

5.1 Predictions

```
test_pred <- predict(iris_svm, test_in)
table(test_pred)
```

```
test_pred
      setosa versicolor  virginica
      7         7         7
```

5.2 Results

```
conf_matrix <- confusionMatrix(test_pred, test_truth)
conf_matrix
```

Confusion Matrix and Statistics

	Reference		
Prediction	setosa	versicolor	virginica
setosa	7	0	0
versicolor	0	7	0
virginica	0	0	7

Overall Statistics

```

Accuracy : 1
 95% CI : (0.8389, 1)
No Information Rate : 0.3333
P-Value [Acc > NIR] : 9.56e-11
```

```
Kappa : 1
```

```
McNemar's Test P-Value : NA
```

Statistics by Class:

	Class: setosa	Class: versicolor	Class: virginica
Sensitivity	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000
Pos Pred Value	1.0000	1.0000	1.0000
Neg Pred Value	1.0000	1.0000	1.0000
Prevalence	0.3333	0.3333	0.3333
Detection Rate	0.3333	0.3333	0.3333
Detection Prevalence	0.3333	0.3333	0.3333
Balanced Accuracy	1.0000	1.0000	1.0000

The result is 100% accuracy

5.3 Overfitting?

Did the model overfit? even though we got 100% accuracy that might not mean overfitting because:

- setosa is completely linearly separable.
- versicolor vs. virginica are also almost linearly separable in petal space.

6 Train the dataset on radial kernel

- Radial kernel - allows complex curved boundaries
- High cost - tries to classify training points almost perfectly (risk of overfitting)

```
set.seed(42)

iris_svm2 <- svm(Species ~ ., train, kernel = "radial", scale = TRUE, cost = 100)
summary(iris_svm2)
```

Call:

```
svm(formula = Species ~ ., data = train, kernel = "radial", cost = 100,
     scale = TRUE)
```

Parameters:

```
  SVM-Type:  C-classification
 SVM-Kernel:  radial
      cost:   100
```

Number of Support Vectors: 29

```
( 6 11 12 )
```

Number of Classes: 3

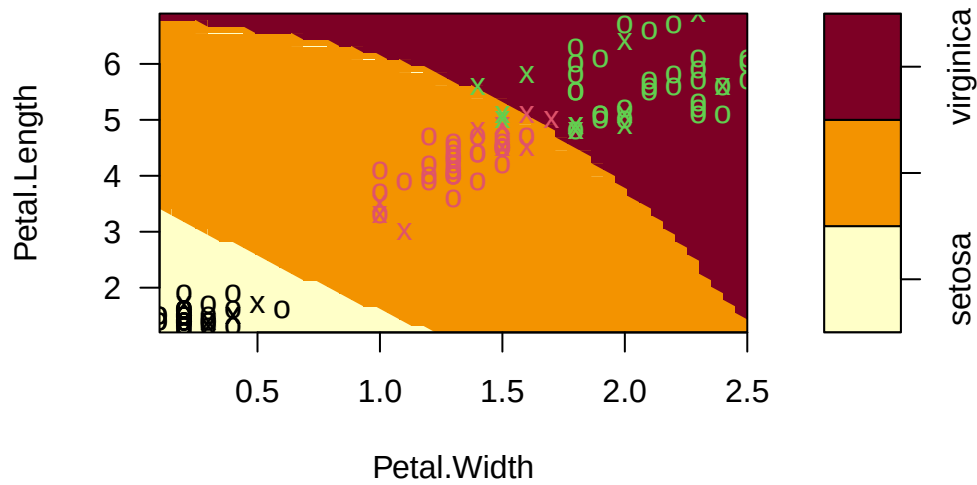
Levels:

```
setosa versicolor virginica
```

6.1 Plots

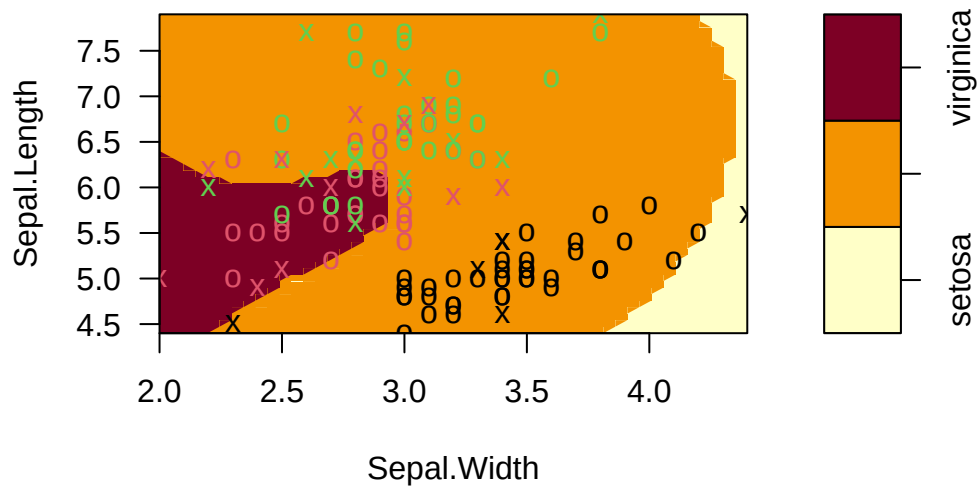
```
plot(iris_svm2, train, Petal.Length ~ Petal.Width, slice = list(Sepal.Length = 6, Sepal.Width = 6))
```

SVM classification plot



```
plot(iris_svm2, train, Sepal.Length ~ Sepal.Width, slice = list(Petal.Length = 4.5, Petal.W
```

SVM classification plot



6.2 Predictions

```
test_pred2 <- predict(iris_svm2, test_in)
table(test_pred2)
```

```
test_pred2
      setosa versicolor  virginica
      7         8         6
```

6.3 Results

```
conf_matrix2 <- confusionMatrix(test_pred2, test_truth)
conf_matrix2
```

Confusion Matrix and Statistics

	Reference		
Prediction	setosa	versicolor	virginica
setosa	7	0	0
versicolor	0	7	1
virginica	0	0	6

Overall Statistics

Accuracy : 0.9524
95% CI : (0.7618, 0.9988)
No Information Rate : 0.3333
P-Value [Acc > NIR] : 4.111e-09

Kappa : 0.9286

McNemar's Test P-Value : NA

Statistics by Class:

	Class: setosa	Class: versicolor	Class: virginica
Sensitivity	1.0000	1.0000	0.8571
Specificity	1.0000	0.9286	1.0000
Pos Pred Value	1.0000	0.8750	1.0000
Neg Pred Value	1.0000	1.0000	0.9333
Prevalence	0.3333	0.3333	0.3333
Detection Rate	0.3333	0.3333	0.2857
Detection Prevalence	0.3333	0.3810	0.2857
Balanced Accuracy	1.0000	0.9643	0.9286

Setosa (perfect): Prediction = Truth in all 7 cases → flawless.

Versicolor (1 mistake): One virginica was misclassified as versicolor.

Virginica (1 mistake): The same misclassification reflects here → 6/7 correct.

Cost (C) controls how strictly the SVM tries to separate the classes.

6.4 High cost (C = large)

Means:

- Misclassification is heavily punished
- SVM tries very hard to separate data perfectly

- Margin becomes narrow
- Only the critical points (right on the boundary) stay as support vectors
- Fewer points are allowed inside the margin Results in fewer support vectors Because the model becomes more rigid and pushes as many points as possible away from the margin.

6.5 Low cost (C small)

Means:

- Misclassification is acceptable
- SVM allows violations
- Margin becomes wide
- More points fall inside or on the margin
- More points become support vectors

Result in more support vectors Because the model becomes more tolerant, allowing many points to influence the boundary.